Responsibly Deploy AI in Quantitative Social Research

As a part of module Social Research Methods

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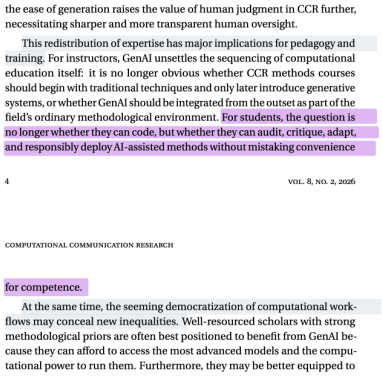
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Using causal and computational methods to evaluate
societal and political impacts of digital information technologies

November 5, 2026; PKU

The goal of this session

(... lower barrier (of computational skills) ... redistribution of expertise ...)

"For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience" (Zhou et al., 2026)



the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

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COMPUTATIONAL COMMUNICATION RESEARCH

for competence.

At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

Source: Zhou et al. (2026)

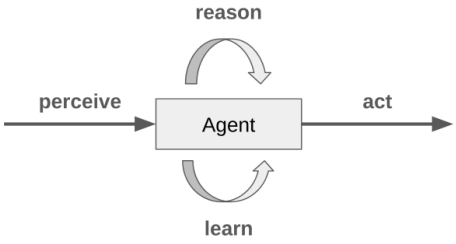
Introduction

1.1 What Is artificial intelligence?

- Computational agents that act intelligently

  (Poole & Mackworth, 2023)

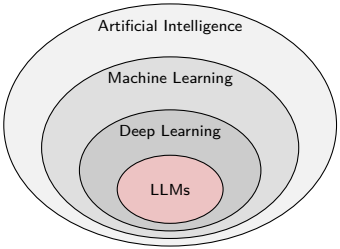
- But what is intelligence – see Figure



Source: Stanford CS221, Percy Liang

1.1 What Is artificial intelligence?

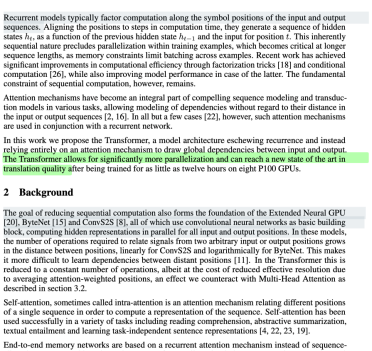
- Computational agents that act intelligently (Poole & Mackworth, 2023);
- But what is intelligence – see Figure
- More importantly, please be ware of the differences ...



1.2 A simplified history of LLMs

- * Artificial intelligence is an old field developed decades ago (see Poole & Mackworth, 2023);
- * Pre-LLMs, early tools mainly classified, clustered, or predicted from human-produced traces (Zhou et al., 2026). [a]
- * Birth of LLMs, Transformers (Vaswani et al., 2017), a model structure to parallelize training enabling large datasets and model sizes
- Post-LLMs, we focus on the LLM era post-2020 (marked by GPT-3; Xu et al., 2026) and its implications for computational social science research; with just a brief tutorical of text classficiation using bert

[a] Likewise, Percy Liang uses 2016, 2018, and Today to divide three time periods, see https://cs336.stanford.edu/lectures/?trace=lecture_01



Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

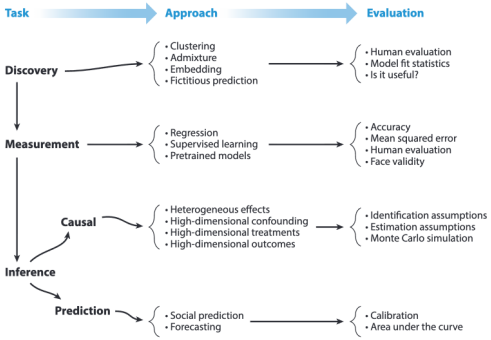
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

Excerpt from Vaswani et al. (2017)

Pre-LLMs AI

2.1 The use of AI in social science before LLMs

- Social science research often leverages machine learning for various tasks, according to the typology in Grimmer et al. (2021):
  - Discovery; e.g., text classification (Tutorial I)
  - Measurement; e.g., regression
  - Causal Inference; e.g., heterogeneous treatment effects
  - Prediction; e.g., forecasting social trends
- Take FIN528 Machine Learning for Finance / Advanced Machine Learning at PHBS if interested



Source: Grimmer et al. (2021) (p.23)

*TUT1

Sentiment classification using RoBERTa

Online material:
Google Colab Link

LLMs

3.1 Roles of LLMs in Computational Social Science

- Large language models (LLMs) are language models roughly with no less than 1 billion parameters (Percy Liang; e.g., GPT, Claude)[1]
- A recent review article suggests four roles of LLMs in computational social science: Subject, assistant, collaborator, evaluator

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

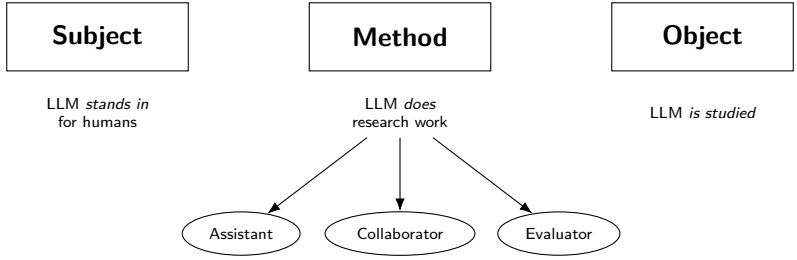
Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

Source: Xu et al. (2026)

[1] We are not a CS class, so it should be fine to use this definition.

3.1 Roles of LLMs in Computational Social Science

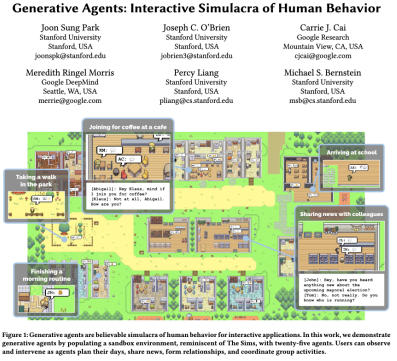
- From more perspectives, we may propose a comprehensive categorization of LLM roles in CSS; to be elaborated in the following slides



Source: Four roles (subjects, assistants, collaborators, evaluators) categorized in Xu et al. (2026); four categories in Hai Liang (2026); and two roles (a research tool, an object of inquiry) adapted from Zhou et al. (2026)

3.2 LLMs as Subjects: Categories

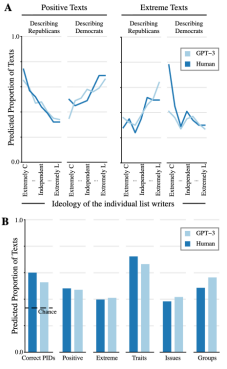
- LLMs serve as research subjects, according to Xu et al. (2026):
  - Generate survey responses (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - Simulating behavior sequences (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - Simulating social interactions (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



Generative agents in a simulated town; Park, O'Brien, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

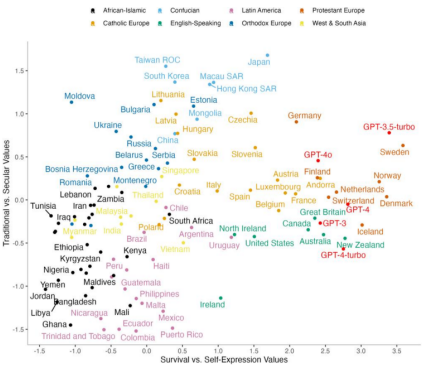
- **Pros:**
  - Successfully replicate human responses in surveys (Argyle, Busby, et al., 2023)
  - Produce believable individual and emergent social behaviors (Park, O'Brien, et al., 2023)
  - Replicate emergence of social conventions (Ashery et al., 2025)
  - etc.



Analysis of GPT-3 and human responses from the Lucid su; Fig 3 in Argyle, Busby, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

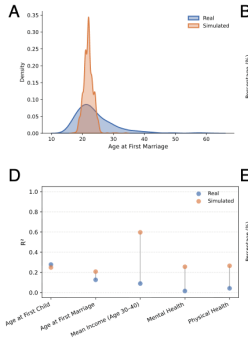
- **Cons:**
  - Poorly refelcting "hard-to-reach populations" (Bollen et al., 2026); fail to capture experiential and cultural depth (Pataranutaporn et al., 2025)
  - Reflect certain partisan (Santurkar et al., 2023) or cultural biases (Tao et al., 2024)
  - Closed to averages but less variation (Bisbee et al., 2024); have strong, varying biases and have low-variance in opinnion or attitudinal research(Boelaert et al., 2025); likely to both misportray and flatten demographic representations (A. Wang et al., 2025); and tend to compress heterogeneity into simplified typological structures (Y. Xie & Xie, 2025)



Deviated LLM-generated answers related to values compared with different countries; Fig 1 in Tao et al. (2024)

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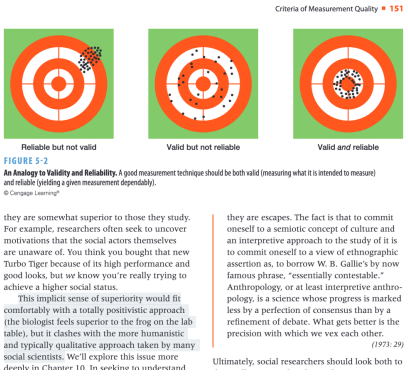
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Failing to capture the dispersion of real-world distribution; Fig 3 in Y. Xie & Xie (2025)

3.4 LLMs as Subjects: Evaluation

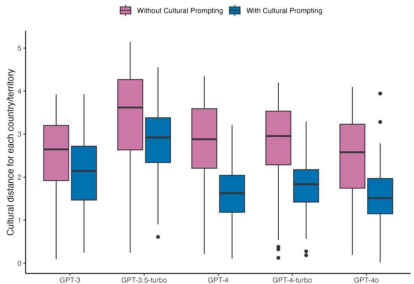
- *Validity* and *reliability* remain concerns (Zhou et al., 2026); please refer to previous lectures by Prof. Ye



Validity and reliability; in Babbie (2021)

3.5 LLMs as Subjects: Improvement

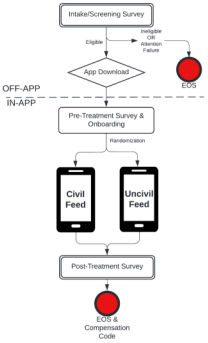
- *Validity* and *reliability* remain concerns (Zhou et al., 2026)
- Three strategies to improve
  - Alignment (pre-use): Adjusting model outputs; mostly done by companies and in computer science studies
  - Optimization (in-use): **Fine-tuning or prompting** strategies to improve performance on specific tasks (e.g., Kolluri et al., 2025; Tao et al., 2024; Y. Xie & Xie, 2025)
  - Correction (post-use): Post-processing corrections ; an emerging promise in econometric and machine learning literature (e.g., Chernozhukov et al., 2018; Egami et al., 2024)



Prompting mitigates cultural biases; Source: Fig 2 in Tao et al. (2024)

3.6 LLMs as Methods: Assistant

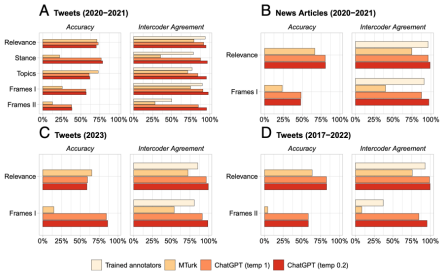
- Interviewing human participants (qualitative)(Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **Supporting experiments** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- Analysing textual content (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Synthetic communicative environment in a randomized experiment; Allamong et al. (2026)

3.6 LLMs as Methods: Assistant

- Interviewing human participants (qualitative)(Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
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- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Better annotation than crowd-workers; Gilardi et al. (2023)

TUT2

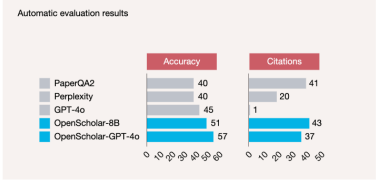
Topic classification using LLMs

Don't forget about validation!

Online material: GitHub

3.7 LLMs as Methods: Collaborator

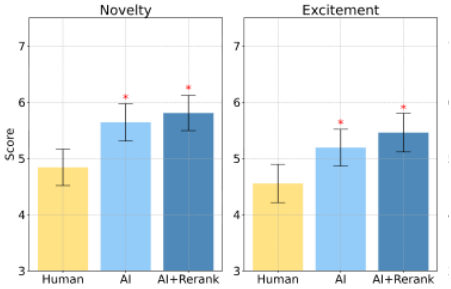
- **Reviewing literature** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- Proposing hypotheses (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)



PhD level literature review system conducted by LLM; see Asai et al. (2026)

3.7 LLMs as Methods: Collaborator

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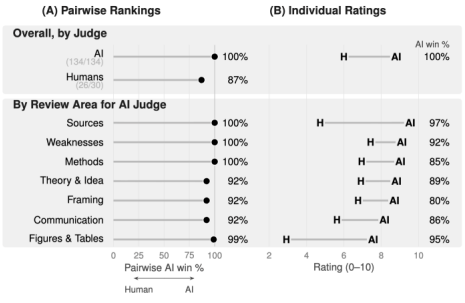
Better novelty and excitement in hypotheses generated by LLM; see Si et al. (2024)

TUT3

Agentic Modelling in Data Analysis
Online material: GitHub

3.8 LLMs as Methods: Evaluator

- Evaluating research protocols (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **Critiquing research outcomes** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; Liang et al., 2024)



AI reviews dominate human reviews; source: (King & Worthington, 2026); not peer-reviewed but well designed

TUT4

Critically evaluate LLM-generated review comments
GitHub

3.9 LLMs as Objects

- "A communication technology that reconfigures production, circulation, interpretation, and power across media systems, raising critical questions about authenticity, access, and the reproduction of social inequalitie." (Zhou et al., 2026)
- For example, Liu and Zhang (2026) evaluates how LLM generated summaries shape public sentiment, the treatment of which is the model.

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

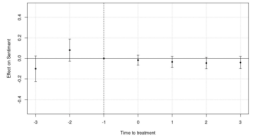


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

AIGS as objects in Liu & Zhang (2026)

Quiz

Propose one research question involving LLMs as objects in the online chatgroup

Please briefly clarify:

(a) the question of interest

(b) operationalization of your variables

(c) the most important figure to be presented

(d) expected contribution

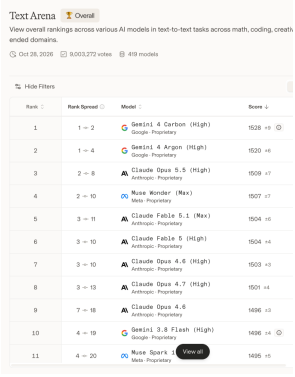
Deploy Responsively

4.1 LLMs deployed responsibly: Selection

- Consider a model as a function, $f(data, algo, object)$, the differences in each part would substantially shape their in-domain capabilities (like better coding or writing)
- In this sense, we may need to choose different models for different tasks, depending on their strengths and limitations

4.1 LLMs deployed responsibly: Selection

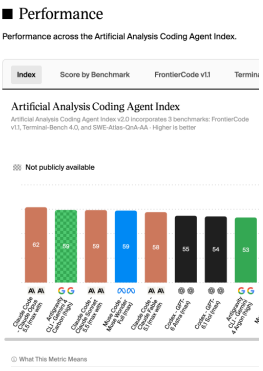
- For example, you may expect Gemini Carbon to excel in writing tasks whereas Claude Opus 5.5 outperforms in engineering



Source: https://arena.ai/leaderboard/chat/text (Nov 3, 2026)

4.1 LLMs deployed responsibly: Selection

- For example, you may expect Gemini Carbon to excel in writing tasks whereas Claude Opus 5.5 outperforms in engineering
- In general, model labs have different commercialization models, for business model optimized for engineering tasks may prioritize coding capabilities (e.g., Anthropic); please check benchmarks / review the literature / do a pilot test before using a model



Source: https://artificialanalysis.ai/agents/coding-agents#coding-agents-index (Nov 3, 2026)

4.2 LLMs deployed responsibly: Validity and Reliability

- Again, always verify the validity and reliability of the model outputs before using them in your research

4.3 LLMs deployed responsibly: Reproducibility

- Reproducibility is important to science
- Ensure that artificial intelligence part of your research is reproducible by
  - Prefer API-based access over chat-based access
  - Documenting model versions, prompts, thinking-level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the seed for randomness and temperature (search for details)

*TUT5 & 6

Multimodal object detection and controlled stimulus generation
Online material: GitHub

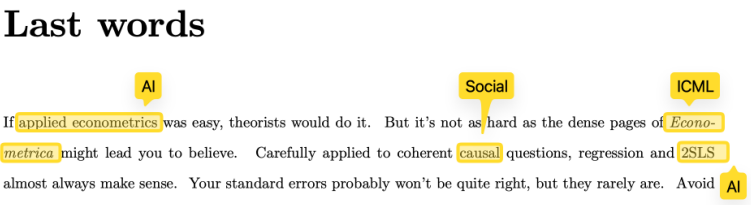
Takeaways

5.1 Takeaways

- AI models (which includes LLMs) are just tools for high-dimensional data analysis; they can be subjects, objects, and methods in quantitative social research
- As methods, they are expected to be used in a responsible way in which validity, reliability, and reproducibility are carefully considered
- Like any others methods, despite they evolve rapidly, their use should be based on our taste for good questions in research, industry, and life, which remain relatively central of higher education.

5.1 Takeaways

- "... Be skeptic... don't panic!" (Angrist & Pischke, 2009)



Source: Angrist & Pischke (2009)

Resources

6.1 Resources

- **Locate the most appropriate AI model for your task**
  - Artificial analysis https://artificialanalysis.ai/
  - Arena.ai, blind benchmarking for AI models https://arena.ai/leaderboard/
- **Find problems in your manuscript**
  - Refine, publication-level technical review; adopted by AEA https://www.refine.ink/
  - OpenAIReview, opensource solution of multi-agent manuscript review by UChicago HAI https://github.com/ChicagoHAI/OpenAIReview
- **Doublecheck references and citations**
  - CiteVerify, "a local tool for verifying references in research papers", https://github.com/cuihuash/CiteVerify
  - Chat with frontier models would also help

6.1 Resources

- **Organize multiple AI models and services**
  - For macOS, magpie, local AI router management for mac https://github.com/yetone/magpie
  - For other platforms, cc-switch, https://github.com/farion1231/cc-switch
- **Read scientific literature using your own language**
  - PDFMathTranslate, read scientific PDFs in your language https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)

Acknowledgments

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References

7.1 References I

Allamong, M. B., Bail, C., Hillygus, D. S., & Volfovsky, A. (2026). The Causal Effects of Political Incivility in Social Media Discussions. Retrieved March 17, 2026, from https://osf.io/download/yaq29

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